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Pelton Wheel Turbine.

The Pelton wheel or Pelton turbine is a tangential flow impulse turbine.

The energy available at the inlet of the turbine is only kinetic energy. The pressure at the inlet and outlet of the turbine is atmospheric. The turbine is used for high heads.

Pelton Wheel at a Hydroelectric Power Plant

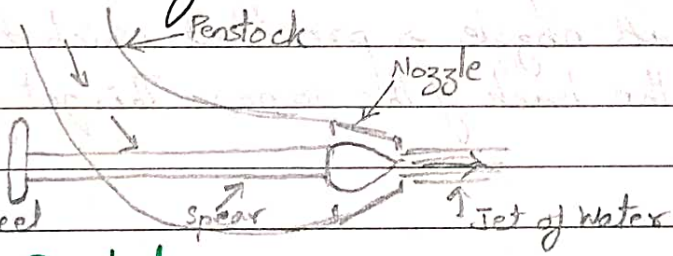
The water from the reservoir flows through the penstock at the outlet of which a nozzle is fitted. The nozzle increases the kinetic energy of the water flowing through the penstock. At the outlet of the nozzle, the water comes out in the form of a jet and strikes the buckets (vanes) of the runner. The main parts of the Pelton turbine are.

→ Nozzle and Flow Regulating Arrangement

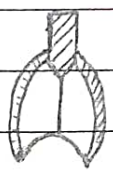
The amount of water striking the buckets (vanes) of the runner is controlled by providing a spear in the nozzle.

The spear is a conical needle which is operated either by a hand wheel or automatically; in an axial direction depending upon the size of the unit.

When spear is pushed forward into the nozzle, the amount of water striking the runner is reduced and vice-versa



→ Runner with Buckets



It consists of a circular disc on the periphery of which a number of buckets, evenly spaced are fixed. The shape of the bucket is of a double hemispherical cup or bowl. Each bucket is divided into two symmetrical parts by a dividing wall, known as splitter.

The jet of water strikes on the splitter. The splitter divides the jet into two equal parts and the jet comes out at the outer edge of the bucket.

The buckets are shaped in such a way that the jet gets deflected through 160° or 170°.

The buckets are made of cast iron, cast steel bronze or stainless steel; depending upon the head at the inlet of the turbine.

→ Casing.

The function of a casing is to prevent the splashing of the water and to discharge water to tail race. It also acts as safegaurd against accidents.

It is made of cast iron or fabricated steel plates. The casing does not perform any hydraulic functions.

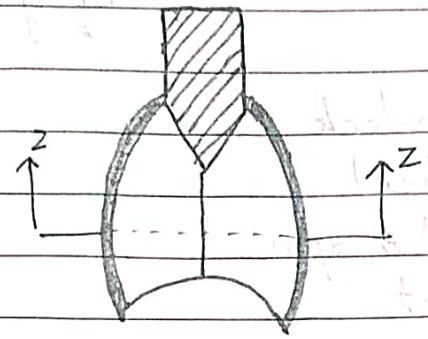
→ Breaking Jet.

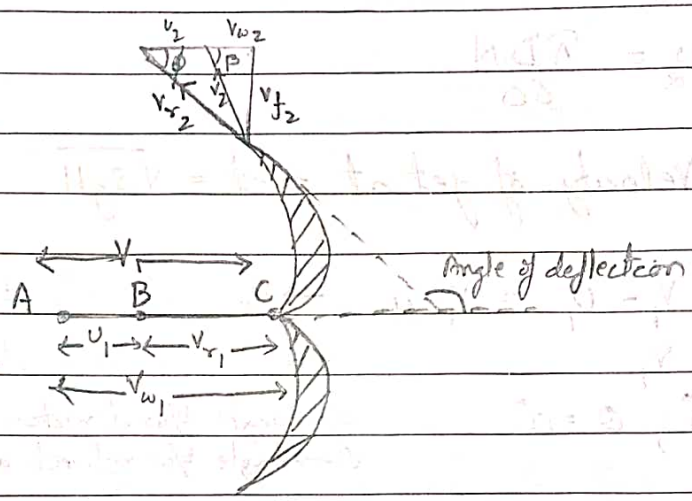
When the nozzle is completely closed by moving the spear in the forward direction, the amount of water striking the runner is reduced to zero.

But, due to inertia, the runner goes on revolving for a long time. To stop the runner in a short time, a small nozzle is provided which directs the jet of water on the back of the vanes. This jet of water is called breaking jet.

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Velocity Triangles and Work done for Pelton Wheel.





The splitter is the inlet tip and outer edge of the bucket is the outlet tip of the bucket.

The inlet velocity triangle is drawn at the splitter and the outlet velocity triangle is drawn at the outer edge of the bucket.

* V_1 → Absolute velocity of jet before striking the bucket
* V_2 → Absolute velocity of jet leaving the bucket.
* U_1 → Absolute velocity of bucket considered along the direction tangential to the pitch circle.

$U_1 = U_2$ as the inlet and outlet tips of the buckets are at the same radial distance, hence the tangential velocity of the bucket at both the tips is same

* V_{r_1} → Relative velocity of the incoming jet
* V_{r_2} → Relative velocity of the jet leaving the bucket
* Actual { V_{w_1}, V_{w_2} → Velocity of whirl at inlet and outlet
  { V_{f_1}, V_{f_2} → Velocity of flow at inlet and outlet

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H = Net head acting on the Pelton wheel.
$= H_g - h_f$

$h_f = \frac{4 f L v^2}{2g D^*}$

$D^* \rightarrow$ Diameter of Penstock
$D \rightarrow$ Diameter of wheel.
$N \rightarrow$ Speed of wheel in rpm
$d \rightarrow$ Diameter of jet.

$$u_1 = u_2 = \frac{\pi D N}{60}$$

Inlet

$V_1 \rightarrow$ Velocity of jet at inlet $= \sqrt{2gH}$

Component of V_1

$V_{r_1} = V_1 - u_1 = V_1 - u \rightarrow$ Relative velocity.
$V_{w_1} = V_1 \rightarrow$ Whirl velocity of absolute at inlet.
$\alpha = 0^\circ, \theta = 0^\circ$
$\alpha \rightarrow$ angle b/w direction of jet & motion (V_1)
$\theta \rightarrow$ angle b/w rel. vel. and motion (V_{r_1})

Outlet

$V_{r_2} = V_{r_1}$
$V_{w_2} = V_{r_2} \cos\phi - u_2 \rightarrow$ whirl velocity of absolute at outlet
↳ component of V_2
$\# \quad V_{f_2} = V_2 \sin\beta$

Force exerted by the jet in the direction of motion
F_x = Mass of water striking per sec × [initial vel. with which jet strikes in direct° of motion − Final vel. of jet in the dir° of motion]

$F_x = \rho a V_1 [V_{w_1} + V_{w_2}]$

$V_{r_1} - V_{r_2}$
$V_1 - u_1 - (-(u_2 + V_{w_2}))$
$V_{w_1} - u_1 + u_2 + V_{w_2}$

This is a case of series of vanes, hence the mass of water striking is $\rho a V_1$ and not $\rho a V_{r_1}$.

a = Area of jet $= \frac{\pi}{4} d^2$

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Work done by the jet on the runner per second
$= F_x \times u$
$= \rho a V_1 [V_{w_1} + V_{w_2}] \times u$ Nm/s

Power given to the runner by the jet
$= \frac{\rho a V_1 [V_{w_1} + V_{w_2}] \times u}{1000}$ kw

Work done/s per unit weight of water striking/s
$= \frac{\rho a V_1 [V_{w_1} + V_{w_2}] \times u}{\text{weight of water striking/s}}$
$= \frac{\rho a V_1 [V_{w_1} + V_{w_2}] \times u}{\rho a V_1 \times g}$
$= \frac{1}{g} [V_{w_1} + V_{w_2}] u$

The energy supplied to the jet at inlet is in the form of kinetic energy and is equal to $\frac{1}{2}mv^2$

∴ K.E. of the jet per second $= \frac{1}{2}(\rho a V_1) \times V_1^2$

Hydraulic Efficiency,

$\eta_h = \frac{\text{wk. done per second}}{\text{K.E. of the jet per second.}}$

$= \frac{\rho a V_1 [V_{w_1} + V_{w_2}] \times u}{\frac{1}{2}(\rho a V_1) V_1^2}$

$$\boxed{\eta_h = \frac{2 [V_{w_1} + V_{w_2}] u}{V_1^2}}$$

Now,

$V_{w_1} = V_1$; $V_{r_1} = V_1 - u$; $V_{r_2} = V_{r_1} = V_1 - u$

$V_{w_2} = V_{r_2} \cos\phi - u_2 = (V_1 - u)\cos\phi - u$

∴ $\eta_h = \frac{2 [V_1 + (V_1 - u)\cos\phi - u] \times u}{V_1^2}$

$$\eta_h = \frac{2(V_1 - u)[1 + \cos\phi]u}{V_1^2}$$

Maximum Efficiency.

The efficiency will be maximum for a given value of V_1, when

$$\frac{d}{du}(\eta_h) = 0$$

$$\frac{d}{du}\left[\frac{2u(V_1 - u)(1 + \cos\phi)}{V_1^2}\right] = 0$$

$$\frac{(1 + \cos\phi)}{V_1^2}\frac{d}{du}(2uV_1 - 2u^2) = 0$$

$$\frac{d}{du}(2uV_1 - 2u^2) = 0 \qquad \because \frac{1 + \cos\phi}{V_1^2} \neq 0$$

$$2V_1 - 4u = 0$$

$$u = V_1/2$$

Hence, the hydraulic efficiency of a Pelton wheel will be maximum when the velocity of the wheel is half the velocity of the jet of water at inlet.

$$\eta_h = \frac{2(V_1 - V_1/2)(1 + \cos\phi)(V_1/2)}{V_1^2}$$

$$\eta_h = \frac{(1 + \cos\phi)}{2}$$

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Conditions for Pelton wheel

1. The velocity of jet at inlet is given by
$$V_1 = C_V \sqrt{2gH}$$
$C_V \rightarrow$ coefficient of velocity = 0.98 or 0.99
$H \rightarrow$ Net head on turbine.

2. The velocity of wheel, u is given by $u = \phi \sqrt{2gH}$
$\phi \rightarrow$ Speed ratio (0.43 to 0.48)

3. The angle of deflection of the jet through buckets is taken as 165°; if no angle of deflection is given.

4. The mean diameter or the pitch diameter, D of the Pelton wheel is given by,
$$u = \frac{\pi D N}{60} \quad \text{or} \quad D = \frac{60u}{\pi N}$$

5. Jet Ratio, is defined as the ratio of the pitch diameter, D of the Pelton wheel to the diameter of the jet, d
$$m = D/d \quad (= 12 \text{ for most cases})$$

6. Number of buckets on a runner is given by
$$Z = 15 + D/2d = 15 + 0.5m$$

7. Number of jets, is obtained by dividing the total rate of flow through the turbine by the rate of flow of water through a single jet.

Width of Bucket = $5d$

Depth of Bucket = $1.2d$

Q1. A pelton wheel has a mean bucket speed of 10 m/s with a jet of water flowing at the rate of 700 lit/s under a head of 30 m. The buckets deflect the jet through an angle of 160°. Calculate the power given by water to the runner and the hydraulic efficiency of the turbine. Assume coefficient of velocity = 0.98.

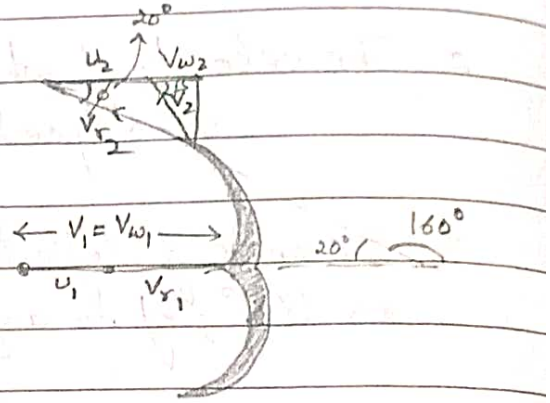
Sol. $\phi = 180° - 160°$; $H = 30\,m$

$= 20°$

$V_1 = C_v\sqrt{2gH}$

$= 0.98 \times \sqrt{2 \times 9.81 \times 30}$

$= 23.77\ m/s.$



$u_1 = u_2 = 10\ m/s$

$Q = 700\ l/s = 0.7\ m^3/s$

$\therefore$ $V_{r_1} = V_1 - u_1 = 23.77 - 10 = 13.77\ m/s$

$V_{w_1} = V_1 = 23.77\ m/s$

$V_{r_2} = V_{r_1} = 13.77\ m/s$

$V_{w_2} = V_{r_2}\cos\phi - u_2 = 13.77\cos 20° - 10 = 2.94\ m/s.$

Work done by the jet per second on the runner

$= \rho a V_1 [V_{w_1} + V_{w_2}] u$

$= 1000 \times 0.7 \times [23.77 + 2.94] \times 10$ $(aV_1 = Q = 0.7\ m^3/s)$

$= 186970\ N\,m/s.$

Power given to turbine $= \frac{186970}{1000} = 186.97\ kW$

Hydraulic Efficiency, $\eta_h = \frac{2[V_{w_1} + V_{w_2}] \times u}{V_1^2}$

$= \frac{2[23.77 + 2.94]10}{23.77^2}$

$= 0.9454$ or 94.54%

Q2. A Pelton wheel is to be designed for the following specifications.

Shaft power = 11,772 kW ; Head = 380 m ; Speed = 750 rpm
Overall Efficiency = 86% ; Jet diameter is not to exceed one-sixth of the wheel diameter.

Determine :-

i) The wheel diameter.
ii) The diameter of the jet
iii) The number of jets required. Take $K_{v_1} = 0.985$, $K_{u_1} = 0.45$

Sol. Given,

Shaft Power = 11,772 kW
$H = 380$ m
$N = 750$ rpm
$\eta_o = 86\%$, 0.86
Ratio of jet dia. to wheel dia. $\Rightarrow d/D = 1/6$
$K_{v_1} = 0.985$ (Coefficient of velocity)
$K_{u_1} = 0.45$ (Speed ratio)

Now,

$$V_1 = C_v\sqrt{2gH} = 0.985 \times \sqrt{2 \times 9.81 \times 380} = 85.05 \text{ m/s}$$

$$u = u_1 = 0.45 \times \sqrt{2 \times 9.81 \times 380} = 38.85 \text{ m/s.}$$

We know,

$$u = \frac{\pi DN}{60} \Rightarrow D = \frac{60u}{\pi N} = \frac{60 \times 38.85}{\pi \times 750}$$

$$\Rightarrow D = 0.989 \text{ m Ans.}$$

$\therefore d = D/6 = 0.165$ m Ans

Now,

$$\eta_o = \frac{S.P}{W.P} = \frac{11772}{\frac{\rho g Q H}{1000}} \qquad \eta_o = 0.86$$

$$\therefore Q = \frac{11772 \times 1000}{0.86 \times 1000 \times 9.81 \times 380} = 3.672 \text{ m}^3/\text{s} \text{ (Total discharge)}$$

Discharge in one jet, $q = \text{Area} \times \text{Vel} = \frac{\pi}{4} \times d^2 \times V_1 = \frac{\pi}{4} \times 0.165^2 \times 85.05$

$= 1.818 \text{ m}^3/\text{s}$.

No. of jets $= Q/q = \frac{3.672}{1.818} = 2$ jets.

Q3. Design a pelton wheel for a head of 80m and speed 300 rpm. The pelton wheel develops 103 kW shaft power. Take $C_v = 0.98$, $\eta_o = 0.8$, ϕ, speed ratio $= 0.45$.

Sol. $V_1 = C_v \sqrt{2gH}$

$= 0.98 \times \sqrt{2 \times 9.81 \times 80} = 38.825$ m/s

$u_1 = u_2 = u = \phi \sqrt{2gH} = 0.45 \times \sqrt{2 \times 9.81 \times 80} = 17.828$ m/s.

$u = \frac{\pi DN}{60}$

$D = \frac{60u}{\pi N} = \frac{60 \times 17.828}{\pi \times 300} = 1.135$ m

$\eta_o = \frac{SP}{\omega P} = \frac{103}{\frac{\rho g Q H}{1000}} = \frac{103 \times 1000}{1000 \times 9.81 \times Q \times 80} = 0.8$

$Q = 0.164$ m³/s.

$Q = aV_1 \Rightarrow 0.164 = \frac{\pi}{4} d^2 \times 38.825$

$\Rightarrow d = 73.37$ mm $\cong 75$ mm

* Width of bucket $= 5d = 5 \times 75 = 375$ mm
* Depth of bucket $= 1.2d = 1.2 \times 75 = 90$ mm.
* No. of buckets on the wheel, $Z = 15 + \frac{D}{2d}$

$Z = 15 + \frac{1.135}{2 \times \frac{75}{1000}} = 22.56 \simeq 23$ Nos

Q4. Two jets strike the buckets of a Pelton wheel, which is having shaft power as 14,715 kW. The diameter of each jet is given as 150 mm. If the net head on the turbine is 500 m, find the overall efficiency of the turbine. $C_v = 1$

Sol. $q = \frac{\pi}{4} d^2 \times V_1$

$= \frac{\pi}{4} \times 0.15^2 \times 99.045$

$= 1.75 \ m^3/s$

$V_1 = C_v \sqrt{2gH}$

$= 1 \times \sqrt{2 \times 9.81 \times 500}$

$= 99.045$

Total discharge $= 2 \times q$

$= 3.5 \ m^3/s$

$$\eta_o = \frac{SP}{WP} = \frac{14715}{\frac{\rho g Q H}{1000}} = \frac{14715 \times 1000}{1000 \times 9.81 \times 3.5 \times 500}$$

$= 0.857$

$= 85.7\%$

Q5. The penstock supplies water from a reservoir to the Pelton wheel with a gross head of 500 m. One third of the gross head is lost in friction in the penstock. The rate of flow of water through the nozzle fitted at the end of the penstock is 2 m³/s. The angle of deflection of the jet is 165°. Determine the power given by the water to the runner and also hydraulic efficiency of the Pelton wheel. $\phi = 0.45$, $C_v = 1$.

Sol. $H_g = 500 \ m$, $h_f = 500/3 = 166.7 \ m$

$H = H_g - h_f = 333.3 \ m$.

$Q = 2 \ m^3/s$.

Angle of deflection $= 165°$

$\alpha = 180° - 165° = 15°$

$V_1 = C_v \sqrt{2gH} = 1 \times \sqrt{2 \times 9.81 \times 333.3} = 80.86 \ m/s$

$u = u_1 = u_2 = 0.45 \times \sqrt{2 \times 9.81 \times 333.3} = 36.387 \ m/s.$

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$V_{r_1} = V_1 - u_1$

$= 44.473 \text{ m/s}$

$V_{w_1} = V_1 = 80.86 \text{ m/s}$

Now,

$V_{r_2} = V_{r_1} = 44.473 \text{ m/s}$

$V_{w_2} = V_{r_2} \cos 15° - u_2$

$= 44.473 \cos 15° - 36.387$

$= 6.57 \text{ m/s}.$

Work done by the jet on the runner,

$\rho a V_1 [V_{w_1} + V_{w_2}] \times u$

$\rho Q [V_{w_1} + V_{w_2}] u$

$1000 \times 2 \times [80.86 + 6.57] 36.387$

$6362630 \text{ Nm/s}.$

Power given by the water to the runner in kW

$= \frac{\text{Wk done per second}}{1000}$

$= \frac{6362630}{1000} = 6362.63 \text{ kW}$

Hydraulic Efficiency,

$\eta_h = \frac{2 [V_{w_1} + V_{w_2}] u}{V_1^2}$

$= \frac{2 [80.86 + 6.57] 36.387}{80.86^2}$

$= 0.9731$

$= 97.31\%$

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Francis Turbine.

Inward Flow Reaction turbine, ~~having radial discharge at outlet is known as Francis Turbine.~~ in which the water enters the runner radially at its outer periphery and leaves axially at its centre is known as Francis Turbine. Francis Turbine is a Mixed-Flow type turbine.

The velocity triangle at inlet and outlet of the Francis turbine are drawn in the same way as in case of inward flow reaction turbine.

The discharge is radial at outlet, the velocity of whirl at outlet (i.e V_{w_2}) will be zero.

Work done by water on the runner per second
$$= \rho a V_1 [V_{w_1} u_1]$$

Work done per second per unit weight of water striking /s.
$$= \frac{1}{g} [V_{w_1} u_1]$$

Hydraulic Efficiency, $\eta_h = \frac{V_{w_1} u_1}{gH}$

1. The ratio of width of the wheel to its diameter, $n = B_1/D_1$
n varies from 0.10 to 0.40

2. Flow ratio $= \frac{V_{f_1}}{\sqrt{2gH}}$ and varies from 0.15 to 0.30

3. Speed ratio $= \frac{u_1}{\sqrt{2gH}}$ and varies from 0.6 to 0.9

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Axial Flow Reaction Turbine

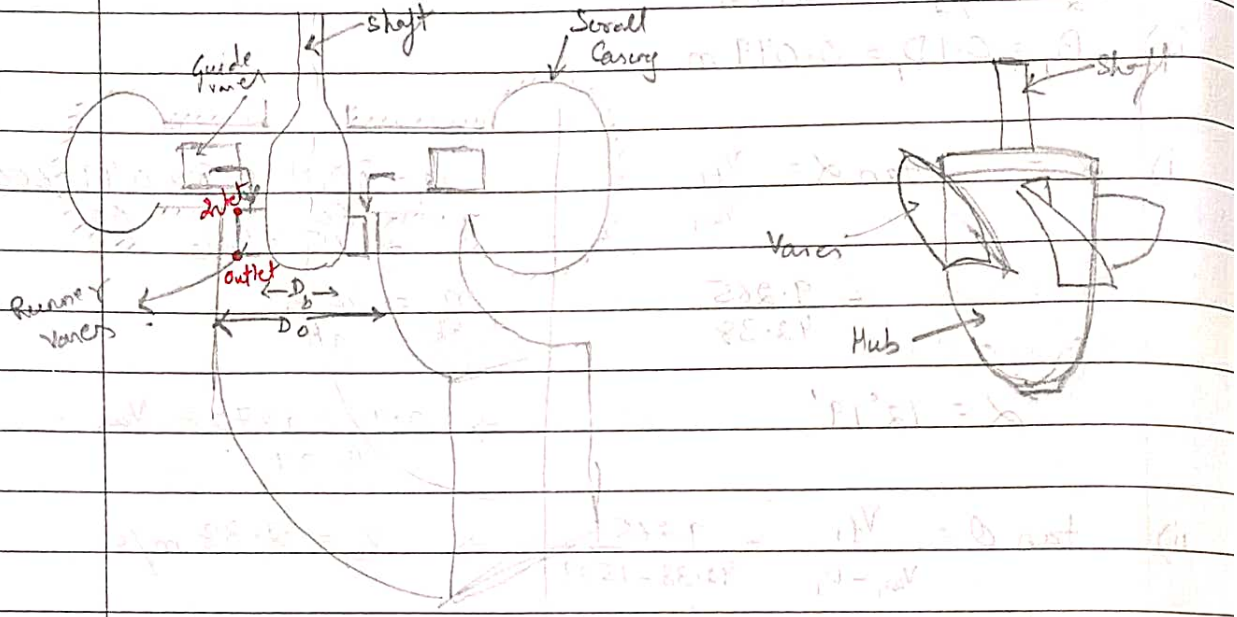
If the water flows parallel to the axis of the rotation of the shaft, the turbine is known as axial flow turbine.

→ If the head at the inlet of the turbine is the sum of pressure energy and kinetic energy (and) during the flow of water through runner a part of pressure energy is converted into kinetic energy, the turbine is known as reaction turbine.

Types of Axial flow reaction turbine

i) Propeller Turbine – Vanes fixed to hub & not adjustable

ii) Kaplan Turbine – Vanes are adjustable.



The main parts of a Kaplan turbine

→ Scroll casing

→ Guide Vane mechanism

→ Hub/Boss with vanes / Runner of the turbine

→ Draft Tube.

* Water from penstock enters the scroll casing and then moves to the guide vanes. From the guide vanes, the water turns 90° and flows axially through the runner.

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Discharge through runner, $Q = \pi/4 (D_o^2 - D_b^2) \times V_{f_1}$

$D_o \rightarrow$ Outer diameter of runner

$D_b \rightarrow$ Diameter of hub/boss

$V_{f_1} \rightarrow$ Velocity of flow at inlet

The peripheral velocity at inlet and outlet are equal.

$$u_1 = u_2 = \frac{\pi D_o N}{60}$$

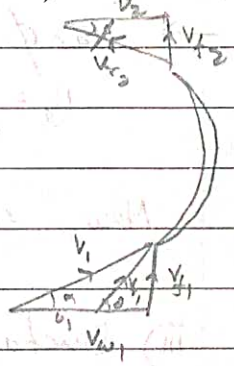
Velocity of flow at inlet and outlet are equal

$$V_{f_1} = V_{f_2}$$

Area of flow at inlet = Area of flow at outlet = $\pi/4 (D_o^2 - D_b^2)$

Q14. A Kaplan turbine working under a head of 20m develops 11772 kw shaft power. The outer diameter of the runner is 3.5m and hub diameter is 1.75m. The guide blade angle at the extreme edge of the runner is 35°. The hydraulic and overall efficiencies of the turbines are 88% and 84% respectively. If the velocity of whirl is zero at outlet, determine
i) Runner vane angles at inlet and outlet at the extreme edge of the runner
ii) Speed of the turbine.

Sol. $H = 20m$, $S.P = 11772$ kw, $D_o = 3.5m$, $D_b = 1.75m$, $\alpha = 35°$,
$\eta_h = 88\%$, $\eta_o = 84\%$, $V_{w_2} = 0$



$$\rightarrow \eta_o = \frac{S.P}{W.P} \Rightarrow 0.84 = \frac{11772 \times 1000}{\rho g Q H}$$

$$\Rightarrow Q = 71.428 \ m^3/s.$$

$$\rightarrow Q = \pi/4 (D_o^2 - D_b^3) V_{f_1}$$

$$\Rightarrow V_{f_1} = 9.9 \ m/s.$$

$\alpha = 35°$

$$\rightarrow \tan\alpha = \frac{V_{f_2}}{V_{w_1}} \Rightarrow V_{w_1} = 14.14 \ m/s$$

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Draft-Tube

A draft tube is a pipe of gradually increasing area which connects the outlet of the runner to the tail race. It has the following two purposes (+ passage for water discharge)

i) It permits a negative head to be established at the outlet of the runner and thereby increase the net head on turbine.

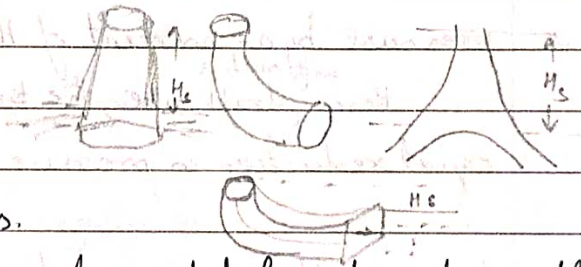
→ The turbine may be placed above the tail race without any loss of net head and hence turbine may be inspected properly.

ii) It converts a large proportion of K.E ($V_2^2/2g$) rejected at the outlet of the turbine into useful pressure energy.

→ Without the draft tube, K.E rejected at the outlet of the turbine will go waste to the tail race.

Hence, the draft-tube increases the net head on the turbine. The turbine develops more power and also the efficiency of the turbine increases.

Types of Draft-tubes



1. Conical draft tube
2. Simple elbow tubes
3. Moody Spreading tubes.
4. Elbow draft-tube with circular inlet & rectangular outlet.

1, 3 → Most efficient

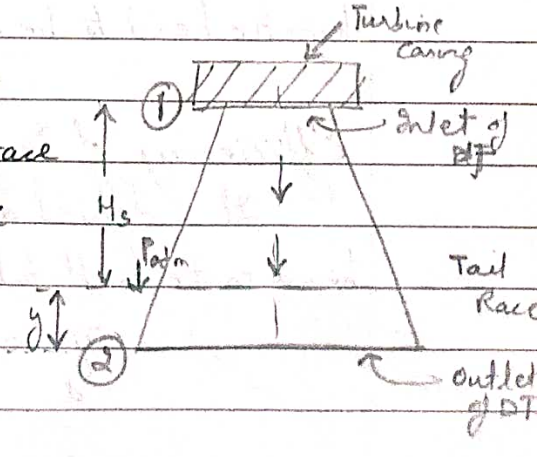
2, 4 → Require less space

Draft tube Theory.

Consider a conical draft-tube

H_s → Vertical ht. of draft-tube above tail race

y → Distance of bottom of dt from tail race



Applying Bernoulli's equation at inlet (1-1) and outlet (2-2) of the draft tube, assuming (2-2) as datum.

$$\frac{P_1}{\rho g} + \frac{V_1^2}{2g} + (H_s + y) = \frac{P_2}{\rho g} + \frac{V_2^2}{2g} + 0 + h_f$$

$h_f \rightarrow$ loss of energy.

But, $\frac{P_2}{\rho g} = \frac{P_a}{\rho g} + y$

$\therefore P_1/\rho g + V_1^2/2g + H_s + y = P_a/\rho g + y + V_2^2/2g + h_f$

$$P_1/\rho g = P_a/\rho g - H_s - \left(\frac{V_1^2}{2g} - \frac{V_2^2}{2g} - h_f\right)$$

Hence, $P_1/\rho g$ is less than atmospheric pressure.

If the draft tube is made cylindrical $\rightarrow V_1 = V_2$
and if frictional loss is neglected $\rightarrow h_f = 0$
$\Rightarrow$ Pressure head at runner exit would be less than atmospheric pressure by an amount of H_s (Ht. of runner above tail race)
Hence, with such a draft tube, the turbine would not loose head H_s because of equal reduction in pressure head at the runner exit.

If draft tube is gradually increasing c/s.
The value of pressure at runner exit is further reduced by an amount equal to $\left(V_1^2/2g - V_2^2/2g - h_f\right)$
In other words, such a draft tube causes a large portion of kinetic head to be converted into pressure head.

Efficiency of Draft-Tube.

It is defined as the ratio of actual conversion of K.E into pressure head in the draft-tube to the kinetic head at inlet of draft tube.

Theoretical conversion of K.E to Pressure head

$$\eta_d = \frac{\left(V_1^2/2g - V_2^2/2g\right) - h_f}{V_1^2/2g}$$

η_b